



## **COMPLEX ENGINEERING PROBLEM-REPORT**

**TITLE:** DEVELOPMENT OF REINFORCED CONCRETE PROBLEM  
USING PYTHON CODE

**COURSE:** REINFORCED CONCRETE DESIGN – II

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**DATE:** 17 NOVEMBER, 2025

**PROBLEM CHOSEN:** DESIGN AND DETAILING OF A RECTANGULAR  
COMBINE FOOTING

**SUBMITTED TO:** DR. ASLAM FAQEER MOHAMMAD

**PROBLEM STATEMENT:** The objective of this Complex Engineering Problem (CEP) is to develop a Python based tool that automates the analysis, design, and detailing of a combined footing supporting two columns under given loads. The program is capable of taking user inputs such as column loads, geometry, material properties, and load factors, and then performs: Structural analysis (shear and moment calculations), Flexural design, Reinforcement detailing, Code compliance checks (as per chosen design code). The developed code provides both numerical design outputs and detailing outputs.

## **DESIGN CODES AND ASSUMPTIONS**

### **Design Code Used:**

The design has been carried out as per **ACI 318-19**: Building Code Requirements for Structural Concrete

### **Assumptions:**

Concrete Unit Weight: 150 lb/ft<sup>3</sup>

Soil Unit Weight = 100 lb/ft<sup>3</sup>

Reinforcement Cover: 3 inches

Load Factors:

Dead Load (DL): 1.2

Live Load (LL): 1.6

Columns are rectangular with fixed loads acting concentrically.

The footing rests on uniform soil pressure.

## **DESCRIPTION OF ALGORITHM**

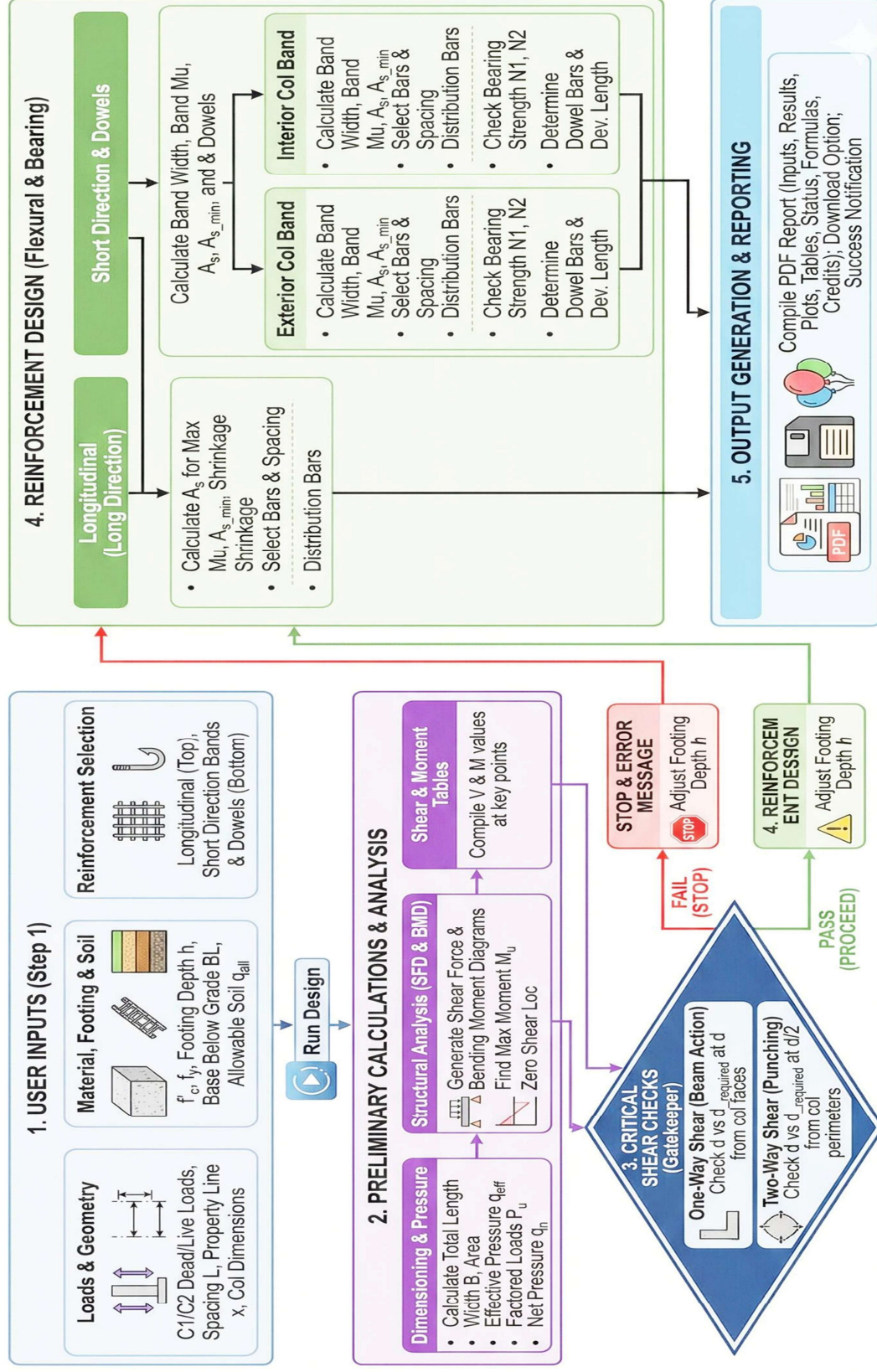
The developed Python program follows a step-by-step procedure for the analysis and design of a combined footing. First, the program accepts user inputs including column loads, spacing, material strengths, footing dimensions, and soil bearing capacity. Using these inputs, it calculates the required footing area and determines soil pressure under service and factored load combinations.

Next, the program performs shear and bending moment analysis along the footing length to identify the critical design sections. Based on the maximum bending moment, the required effective depth and reinforcement area are computed using ACI 318 design equations. The program then checks minimum reinforcement limits, shear capacity, and development length to ensure code compliance.

Finally, the program selects practical reinforcement bar sizes and spacing, generates the detailing layout, and outputs a summary of the design results. All key calculations and code checks are performed automatically to assist with reliable and efficient structural design.

# Rectangular Combined Footing Design App (ACI 318-19) - Workflow & Logic

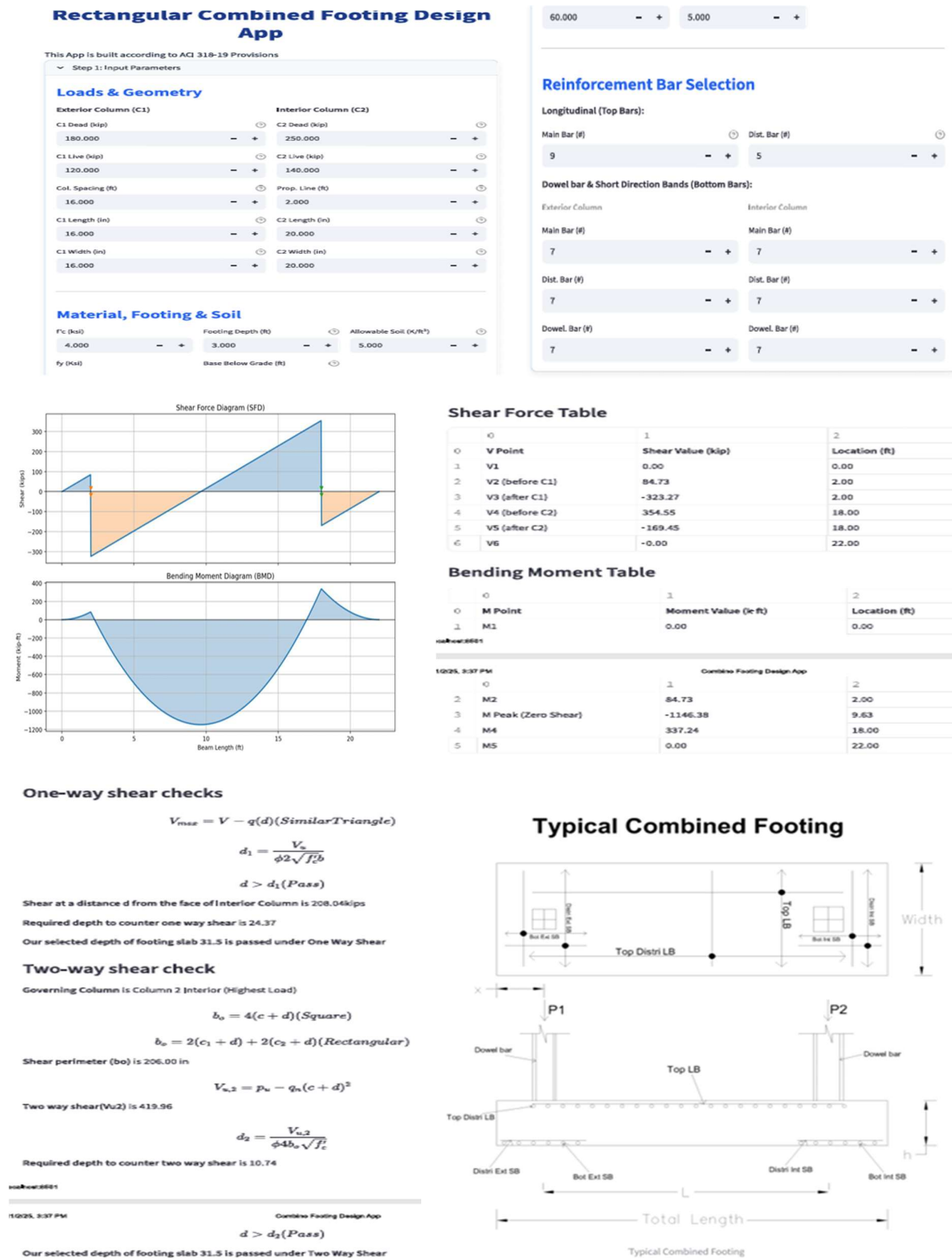
## Flowchart:



Credits: Developed by NED University Students under Faculty Guidance (Dr. Asad-ur-Rehman Khan, Dr. Aslam Fazeer Muhammad)

# DESIGN AND ANALYSIS OF A RECTANGULAR COMBINED FOOTING

## Example 01: Graphical User Interface(Inputs, interpretations and output)



**Python Programming (Input, Interpretations & Results):**

**Rectangular Combined Footing Design Report**

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| Parameter                 | Value |
|---------------------------|-------|
| C1D (kip)                 | 180.0 |
| C2L (kip)                 | 120.0 |
| C2D (kip)                 | 250.0 |
| C2L (kip)                 | 140.0 |
| L (ft)                    | 16.0  |
| x (ft)                    | 2.0   |
| col1l (in)                | 16.0  |
| col1w (in)                | 16.0  |
| col2l (in)                | 20.0  |
| col2w (in)                | 20.0  |
| f'c (ksi)                 | 4.0   |
| fy (K/in <sup>2</sup> )   | 60.0  |
| h (ft)                    | 3.0   |
| qall (K/in <sup>2</sup> ) | 5.0   |
| BL (ft)                   | 5.0   |

| Result                        | Value  |
|-------------------------------|--------|
| Total length (ft)             | 22     |
| qeff (K/ft <sup>2</sup> )     | 5.648  |
| Footing Width (ft)            | 7.5    |
| Total Area (ft <sup>2</sup> ) | 165.0  |
| C1_factored (kip)             | 408.00 |
| C2_factored (kip)             | 524.00 |
| Selected depth (in)           | 31.500 |

**Shear Checks**

"Shear at a distance d from the face of Interior Column" is 208.04kips  
"Required depth to counter one way shear" is 24.37  
"Our selected depth of footing slab 31.5 is passed under One Way Shear"

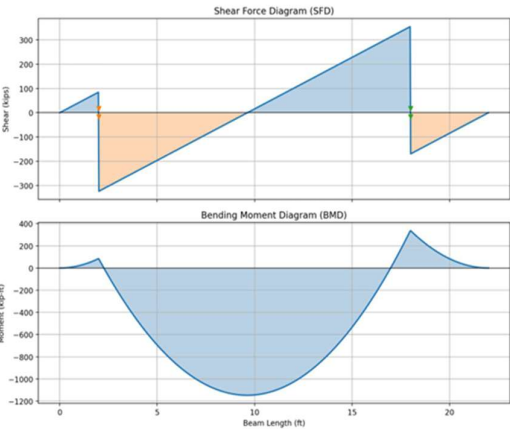
Governing Column: Column 2 Interior (Highest Load)  
Shear perimeter (bo) is 206.00 in  
Two way shear (Vu2) is 419.96  
Required depth to counter two way shear is 10.74

**Shear Force Table**

| **V Point**    | **Shear Value (kip)** | **Location (ft)** |
|----------------|-----------------------|-------------------|
| V1             | 0.00                  | 0.00              |
| V2 (before C1) | 84.73                 | 2.00              |
| V3 (after C1)  | -323.27               | 2.00              |
| V4 (before C2) | 354.55                | 18.00             |
| V5 (after C2)  | -169.45               | 18.00             |
| V6             | -0.00                 | 22.00             |

**Bending Moment Table**

| **M Point**         | **Moment Value (k-ft)** | **Location (ft)** |
|---------------------|-------------------------|-------------------|
| M1                  | 0.00                    | 0.00              |
| M2                  | 84.73                   | 2.00              |
| M Peak (Zero Shear) | -1146.38                | 9.63              |
| M4                  | 337.24                  | 18.00             |
| M5                  | 0.00                    | 22.00             |



**Flexural Reinforcement Summary**

| Item                          | Value       |
|-------------------------------|-------------|
| Mu (k-ft)                     | -1146.38    |
| As (in <sup>2</sup> )         | 8.30        |
| As shrink (in <sup>2</sup> )  | 5.83        |
| As min (in <sup>2</sup> )     | 9.45        |
| Longitudinal bar #            | 9           |
| Number of bars                | 10          |
| Spacing (in)                  | 9           |
| Top LB, Placement             | 10#9@9"c/c  |
| Distribution bar #            | 5           |
| Distribution Number of bars   | 20          |
| Distribution bar Spacing (in) | 13          |
| Top Distri LB, Placement      | 20#5@13"c/c |
| Development Length ld (in)    | 70          |

**Short Direction Reinforcement**

| Item                          | Value      | Item                          | Value      |
|-------------------------------|------------|-------------------------------|------------|
| Bandwidth (ft)                | 5.50       | Band1 (ft)                    | 7.00       |
| Mu (k-ft)                     | 258.59     | Mu (k-ft)                     | 266.16     |
| As (in <sup>2</sup> )         | 1.84       | As (in <sup>2</sup> )         | 1.89       |
| As shrink (in <sup>2</sup> )  | 4.28       | As shrink (in <sup>2</sup> )  | 5.44       |
| As min (in <sup>2</sup> )     | 6.93       | As min (in <sup>2</sup> )     | 8.82       |
| Bar #                         | 7          | Bar #                         | 7          |
| No of Bars                    | 12         | Number of Bars                | 15         |
| Spacing (in)                  | 5          | Spacing (in)                  | 5          |
| Bot Ext SB, Placement         | 12#7@5"c/c | Bot Int SB, Placement         | 15#7@5"c/c |
| Distribution bar #            | 7          | Distribution bar #            | 7          |
| Distribution Number of bars   | 8          | Distribution Number of bars   | 10         |
| Distribution bar Spacing (in) | 8          | Distribution bar Spacing (in) | 8          |
| Distri Ext SB, Placement      | 8#7@8"c/c  | Distri Int SB, Placement      | 10#7@8"c/c |
| Dowel ld (in)                 | 17.00      | Dowel ld (in)                 | 17.00      |



## Manual Solution:

**Solved Example 1:**

**Data:**

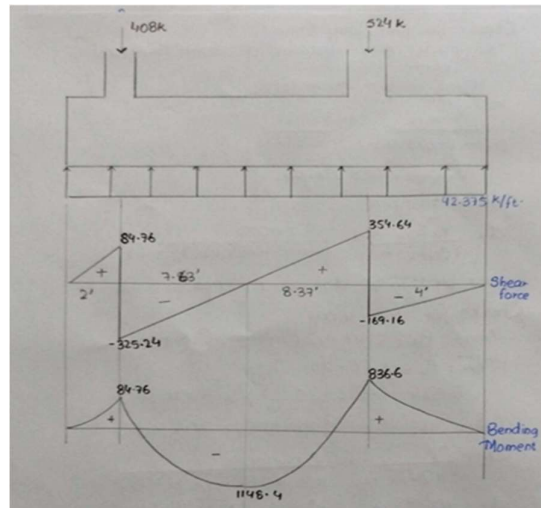
$C_{ext} = (16 \times 16) \text{ inch}^4$      $C_{int} = (20 \times 20) \text{ inch}^4$

$DL_{ext} = 280 \text{ kips}$      $DL_{int} = 250 \text{ kips}$

$L.L._{ext} = 120 \text{ kips}$      $L.L._{int} = 140 \text{ kips}$

Depth below final Grade: 5 ft.

$q_s = 5 \text{ k/ft}^2$      $f'_c = 4 \text{ ksi}$      $f_y = 60 \text{ ksi}$



**Check for One way Shear:**  
 $V_u$  occur @ a distance 'd' from Interior Column.  
 Using Similar triangle  

$$\frac{V_{u \max}}{(8.37 \times 12 - 10 \times 3.5)} = \frac{354.84}{8.37 \times 12}$$
 $V_{u \max} = 208.2 \text{ kips}$   
**Calculating 'd'**  

$$d = \frac{V_{u \max}}{2 \phi F_y \times b} = \frac{208.2 \times 1000}{2(0.75)(4000)(7.5 \times 12)}$$
 $d = 24.4 \text{ in} < \text{assumed (pass)}$   
**Check for two way:**  
 Max  $P_u = 524 \text{ kips}$  (Interior Column).  
 $V_{u1} = P_u - q_u (C + d)^2$   
 $= 524 - 5.65 \left( \frac{20}{12} + \frac{31.5}{12} \right)^2 = 420 \text{ kips}$   
 $b_o = 4(C + d) = 4(20 + 31.5) = 206 \text{ inch}$   

$$d_1 = \frac{V_{u1}}{4 \phi F_y b_o} = \frac{420 \times 1000}{4(0.75)(4000)(206)}$$
 $d_1 = 10.75 \text{ in} < \text{assumed (Safe)}$

$A_s = 9.45 \text{ inch}^2$   
 Using Area of Bar #9  
 $n = 9.45 / 1 = 9.45 \approx 10 \text{ bars}$   

$$\text{Spacing} = \frac{7.5 - 2(\text{cover})}{n - 1}$$

$$= \frac{7.5 - 2(3)}{10 - 1}$$
 $s = 9 \text{ inches}$   
**Distributing bars:**  
 $A_s = 5.83 \text{ inch}^2$      $A_b = \#5 = 0.306 \text{ inch}^2$   
 $n = \frac{5.83}{0.306} = 19.2 \approx 20 \text{ bars}$   
 (S)

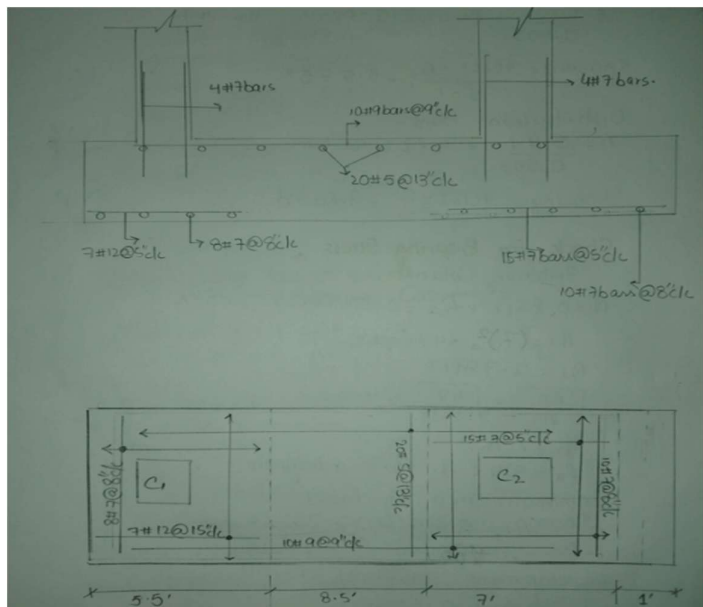
$$\text{Spacing} = \frac{L - 2(\text{cover})}{n - 1}$$

$$= \frac{22(12) - 2(3)}{20 - 1}$$
 $\text{Spacing} = 13 \text{ inches}$

**Area of Steel:**  
 $A_{s1} = 0.001 \times (5.5 \times 12) \times (31.5) = 1.66 \text{ inch}^2$   
 $A_{s2} = \frac{200}{f_y} \times b \times d = \frac{200}{60,000} \times 5.5 \times 12 \times 31.5 = 6.93 \text{ inch}^2$   
 $A_s = 0.0018 \times b \times h = 0.0018 \times 5.5 \times 12 \times 36 = 4.28 \text{ inch}^2$   
**Shrinkage**  
 $A_s = 6.93 \text{ inch}^2$      $A_b = \#7 = 0.60 \text{ inch}^2$   
 $n = \frac{6.93}{0.60} \approx 12 \text{ bars}$   

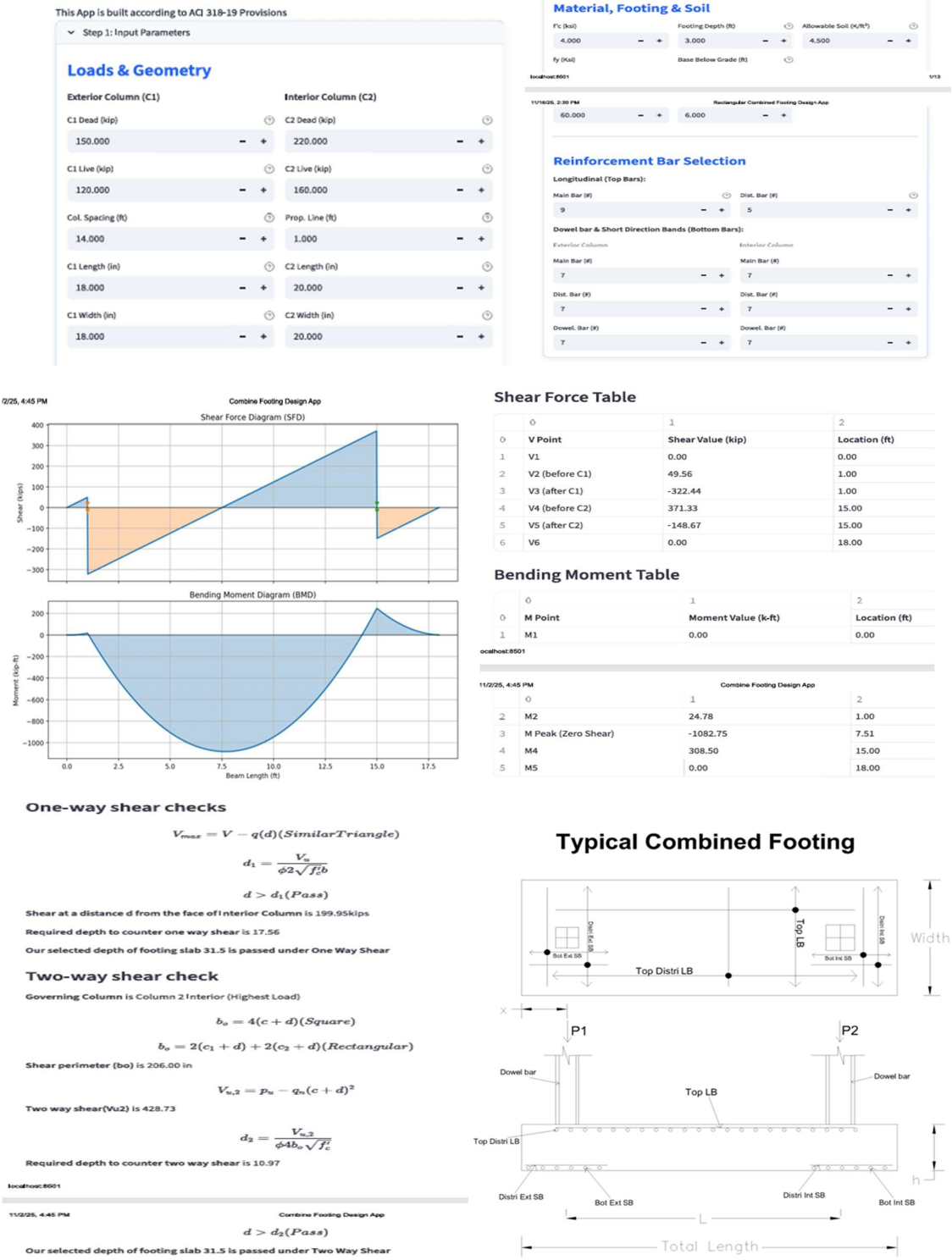
$$\text{Spacing} = \frac{5.5(12) - 3}{12 - 1} = 5"$$
  
**Distribution Bars:**  
 $A_s \text{ shrinkage} = 4.28$      $A_b = 7 = 0.60 \text{ inch}^2$   
 $n = \frac{4.28}{0.60} = 7.1 \approx 8 \text{ bars}$   

$$\text{Spacing} = \frac{5.5(12) - 3}{8 - 1} = 8"$$



Example 02: Graphical User Interface(Inputs, interpretations and output)

Rectangular Combined Footing Design App



**Python Programming (Input, Interpretations & Results):**

**Rectangular Combined Footing Design Report**

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| Parameter                             | Value |
|---------------------------------------|-------|
| C1D (kip)                             | 150.0 |
| C2L (kip)                             | 120.0 |
| C2D (kip)                             | 220.0 |
| C2L (kip)                             | 160.0 |
| L (ft)                                | 14.0  |
| x (ft)                                | 1.0   |
| col1l (in)                            | 18.0  |
| col1w (in)                            | 18.0  |
| col2l (in)                            | 20.0  |
| col2w (in)                            | 20.0  |
| f <sub>c</sub> (ksi)                  | 4.0   |
| f <sub>y</sub> (K/in <sup>2</sup> )   | 60.0  |
| h (ft)                                | 3.0   |
| q <sub>all</sub> (K/in <sup>2</sup> ) | 4.5   |
| BL (ft)                               | 6.0   |

| Result                                | Value  |
|---------------------------------------|--------|
| Total length (ft)                     | 18     |
| q <sub>eff</sub> (K/ft <sup>2</sup> ) | 4.956  |
| Footing Width (ft)                    | 10.0   |
| Total Area (ft <sup>2</sup> )         | 180.0  |
| C1_factored (kip)                     | 372.00 |
| C2_factored (kip)                     | 520.00 |
| Selected depth (in)                   | 31.500 |

**Shear Checks**

"Shear at a distance d from the face of Interior Column" is 199.95kips

"Required depth to counter one way shear" is 17.56

"Our selected depth of footing slab 31.5 is passed under One Way Shear"

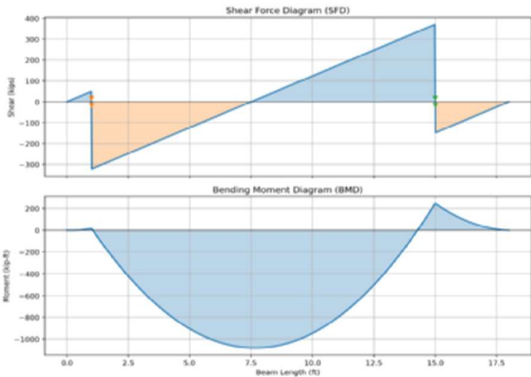
Governing Column: Column 2 Interior (Highest Load)

Shear perimeter (bo) is 206.00 in

Two way shear (Vu2) is 428.73

Required depth to counter two way shear is 10.97

**Shear & Moment Diagrams**



**Shear Force Table**

| "V Point"      | "Shear Value (kip)" | "Location (ft)" |
|----------------|---------------------|-----------------|
| V1             | 0.00                | 0.00            |
| V2 (before C1) | 49.56               | 1.00            |
| V3 (after C1)  | -322.44             | 1.00            |
| V4 (before C2) | 371.33              | 15.00           |
| V5 (after C2)  | -148.67             | 15.00           |
| V6             | 0.00                | 18.00           |

**Bending Moment Table**

| "M Point"           | "Moment Value (k-ft)" | "Location (ft)" |
|---------------------|-----------------------|-----------------|
| M1                  | 0.00                  | 0.00            |
| M2                  | 24.78                 | 1.00            |
| M Peak (Zero Shear) | -1082.75              | 7.51            |
| M4                  | 308.50                | 15.00           |
| M5                  | 0.00                  | 18.00           |

**Flexural Reinforcement Summary**

| Item                          | Value      |
|-------------------------------|------------|
| Mu (k-ft)                     | -1082.75   |
| As (in <sup>2</sup> )         | 7.78       |
| As shrink (in <sup>2</sup> )  | 7.78       |
| As min (in <sup>2</sup> )     | 12.60      |
| Longitudinal bar #            | 9          |
| Number of bars                | 13         |
| Spacing (in)                  | 9          |
| Top LB, Placement             | 13#9@9"c/c |
| Distribution bar #            | 5          |
| Distribution Number of bars   | 26         |
| Distribution bar Spacing (in) | 8          |
| Top Distri LB, Placement      | 26#5@8"c/c |
| Development Length ld (in)    | 70         |

**Short Direction Reinforcement**

| Item                          | Value      | Item                          | Value      |
|-------------------------------|------------|-------------------------------|------------|
| Bandwidth (ft)                | 4.50       | Band1 (ft)                    | 7.00       |
| Mu (k-ft)                     | 335.96     | Mu (k-ft)                     | 264.13     |
| As (in <sup>2</sup> )         | 2.40       | As (in <sup>2</sup> )         | 1.88       |
| As shrink (in <sup>2</sup> )  | 3.50       | As shrink (in <sup>2</sup> )  | 5.44       |
| As min (in <sup>2</sup> )     | 5.67       | As min (in <sup>2</sup> )     | 8.82       |
| Bar #                         | 7          | Bar #                         | 7          |
| No of Bars                    | 10         | Number of Bars                | 15         |
| Spacing (in)                  | 5          | Spacing (in)                  | 5          |
| Bot Ext SB, Placement         | 10#7@5"c/c | Bot Int SB, Placement         | 15#7@5"c/c |
| Distribution bar #            | 7          | Distribution bar #            | 7          |
| Distribution Number of bars   | 6          | Distribution Number of bars   | 10         |
| Distribution bar Spacing (in) | 9          | Distribution bar Spacing (in) | 8          |
| Distri Ext SB, Placement      | 6#7@9"c/c  | Distri Int SB, Placement      | 10#7@8"c/c |
| Dowel ld (in)                 | 17.00      | Dowel ld (in)                 | 17.00      |



## Manual Solution:

**Solved Example 2:**

**Data:**

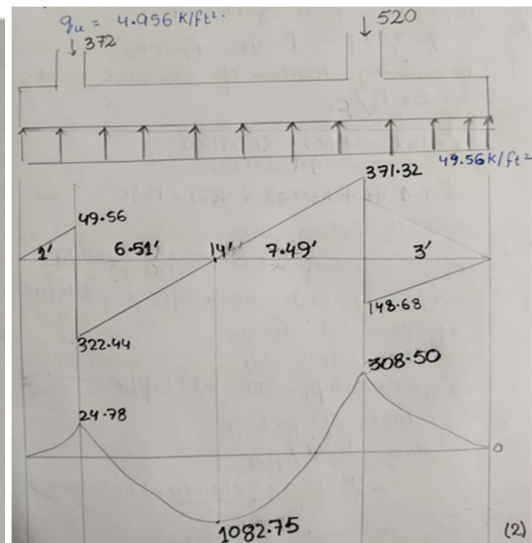
$C_{\text{exterior}} = (18 \times 18) \text{ inch}^2$        $C_{\text{interior}} = (20 \times 20) \text{ inch}^2$

D.L. = 150k      D.L. = 220k

L.L. = 120k      L.L. = 160k

Depth below final grade = 6ft

$q_s = 4.5 \text{ ksf}$        $f'_c = 4 \text{ ksi}$        $f_y = 60 \text{ ksi}$



Check for one way:

$\tau$  occur @ a distance 'd' from Interior Column.

Using Similar Triangle

$$\frac{V_{\text{max}}}{(7.49 \times 12 - 10 - 31.5)} = \frac{371.32}{9.49 \times 12}$$

$V_{\text{max}} = 199.87 \text{ kips}$

Calculating 'd'

$$d = \frac{V_{\text{max}}}{2\phi f'_c b} = \frac{199.87 \times 1000}{2(0.75)\sqrt{4000} \times (10 \times 12)}$$

$d = 17.55" < d_{\text{assumed}} \text{ pass.}$

Check for two way shear.

Max  $P_u = 520 \text{ k}$  (Interior Column)

$$V_u = P_u - q_u (C + d)^2$$

$$= 520 - 4.956 \left( \frac{20}{12} + \frac{31.5}{12} \right)^2$$

$$= 428.718 \text{ kips}$$

$$b_o = 4(C + d) = 4 \left( \frac{20}{12} + \frac{31.5}{12} \right) = 20.6"$$

$$d_s = \frac{V_u}{4\phi f'_c b_o} = \frac{428.718 \times 1000}{4(0.75)(\sqrt{4000})(20.6)}$$

$d_s = 10.96 < d_{\text{assumed}} \text{ (pass)}$

$A_s = 12.6 \text{ inch}^2$

$A_b = \#9 = 1 \text{ inch}^2$

$n = 12.6 / 1 = 12.6 \approx 13 \text{ bars}$

Spacing =  $L - 2 \text{ cover} / n - 1 = 10(12) - 6 / 12$

Spacing = 9" 13 #9 @ 9" c/c.

Distribution bars.

$A_b = \#5 = 0.603$

$n = 7.76 / 0.603 = 25.2 \approx 26 \text{ bars}$

Spacing =  $12(10) - 6 / 25$

$= 4.56 \approx 4 \text{ #6 #5 @ 4" c/c.}$

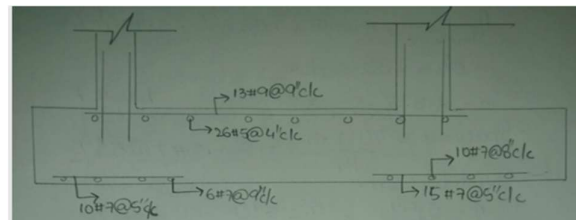
shrinkage

$A_s = 5.67 \text{ inch}^2$

#7 bar

$n = 5.67 / 0.601 = 9.4 \approx 10 \text{ bars}$

Spacing =  $(2(4.5) - 6) / 9 \approx 5" \text{ 10 #7 @ 5" c/c}$



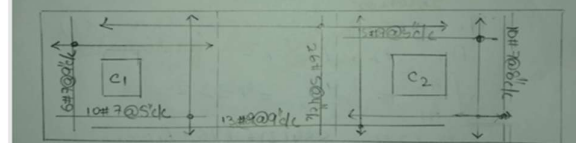
Distribution Bars.

$A_s = 3.4992 \text{ inch}^2$

$n = 3.4992 / 0.601 = 5.8 \approx 6 \text{ bars}$

Spacing =  $\frac{12(4.5) - 6}{5} = 9.6 \approx 9$

6 #7 @ 9" c/c.



**Verification:**

- Hand calculations for moment and soil pressure distribution were performed and matched with the Python output.
- Reinforcement area and effective depth calculations were verified using ACI design equations.
- The results from the program were checked against results from standard reinforced concrete design spreadsheets. The values of bending moment, shear, and reinforcement were found to be very close, which confirms accuracy.
- The program was also tested by changing the loads and footing size. The outputs changed correctly in each case, showing that the code works reliably for different design conditions.

**Possible Extension:**

- Addition of trapezoidal or any other shape combined footing design.
- Integration with AutoCAD or Revit for detailing export.
- The program can be improved to design combined footings where the column loads are not centered, which would lead to uneven soil pressure.
- The program could allow the user to choose different international design codes (such as Eurocode 2 or IS 456) so it can be used in different countries and standards.

**Limitations:**

- Program assumes uniform soil pressure; it does not handle sloped or non-uniform soil conditions.
- Only rectangular combined footings are supported.
- Only for square and rectangular column not for circular column.
- $A_{shrinkage}$  should be calculated according to 60 ksi steel bar.
- No laterals load on columns only vertical loads.
- Only for gravity loads.
- Only concentric loads on column.